

👤 Founder / Team Lead: Utkarsh & Team

IIITA:

☐ Incubated Startup☐ Student Startup☒ Student Innovation☐ Alumni Startup

01

PROBLEM / NEED

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What problem are you solving?

- North-East India suffers frequent, high-fatality landslides every monsoon — roads, homes and lifeline infrastructure are cut off for days.
- Monitoring today is reactive: action is taken only after a slope has already failed, not before.
- One landslide at NHPC's Teesta-V station (Sikkim, Aug 2024) alone caused a loss of Rs 327 crore.
- 329 deaths are recorded in the historical landslide events we studied for this project.
- There is no region-specific, daily-updating early warning system for the North-East today.

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Rs 327 crore lost in a single landslide — NHPC Teesta-V, Sikkim, Aug 2024

02

MOTIVATION / OPPORTUNITY

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Why now: free satellite data + proven ML approach + a validated hardware precedent

- Five satellite missions (rainfall, soil moisture, terrain, vegetation, radar) are now free and queryable in real time via Google Earth Engine.
- NASA's own global warning system, LHASA, proves that satellite data + machine learning is a viable approach at scale.
- IIT Mandi has already proven low-cost ground sensors work in Indian mountain terrain, at ~Rs 1 lakh per site.
- Smart India Hackathon Problem Statement 26001 explicitly calls for exactly this: an AI-based early warning system for NER.
- No existing system combines region-specific ML, live satellite data, and field-level sensing in one platform.



The building blocks already exist: free satellite data, a proven ML approach, a validated low-cost sensor design. No one has combined them for NER yet.

03

PROPOSED SOLUTION

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Brief overview of your product / technology / solution.

- Give SPIREXA any coordinate and any date in the North-East — it returns a landslide risk percentage in about 10 seconds.
- A live GIS dashboard tracks 311 known danger spots across all 8 North-Eastern states, recalculated every day.
- Automated alerts are generated in 9 regional languages, ready for SMS delivery to basic phones.
- An offline-capable field app lets officers report hazards with no network signal.

WHAT WE OBSERVE

Rainfall
NASA GPM — updated every 30 min

Soil moisture
NASA SMAP — has wet the ground to

Slope & terrain
NASA SRTM30 PLUS

Vegetation cover
MODIS 1 km resolution

Ground radar
Sentinel-1 — sees through cloud

Road network
OpenStreetMap

318 past landslides
Historical landslide data

HOW WE COMBINE IT

17 signals per location
- slope steepness > 30° in rain
- rain over 1.5, 3, 7, 15, 30 days
- USGS slope-stability equation
- distance to nearest road

THE AI

Random Forest
- 400 decision trees
- 10-fold cross-validation
- 0.933 accuracy
- 87% of past events caught

WHAT PEOPLE GET

Risk map
Shows roads and villages at risk

Any point, any date
Click to search 25 years

Alerts in 9 languages
SMS ready for basic phones

Works offline
For areas with no signal

Real questions about the model: They are never an input when creating a new location — that is what makes it a prediction rather than a history.

04

HOW IT WORKS

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Explain the working / workflow / architecture.

Satellites + Sensor

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17 Features → ML

→

Risk Score

→

Alert Issued

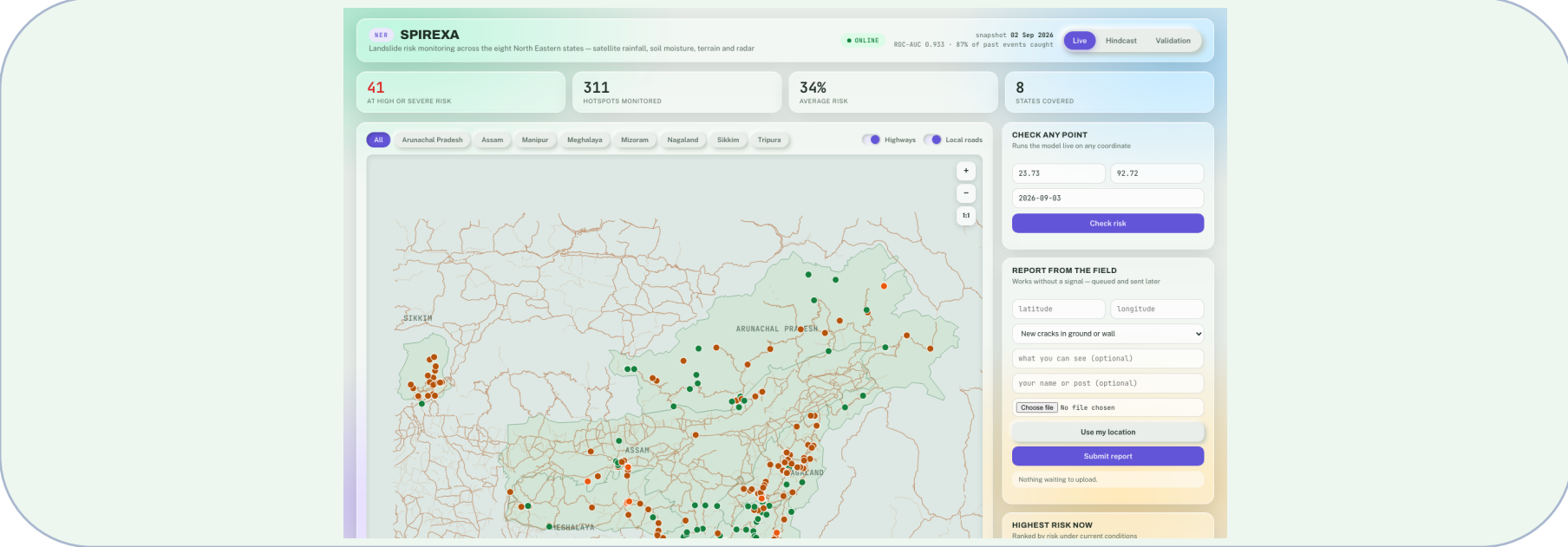
Pipeline: 5 satellites + ground sensor → 17 features → Random Forest → risk % → alert, end-to-end in about 10 seconds.

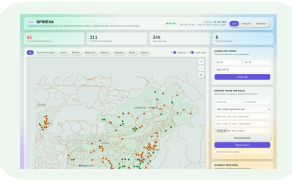
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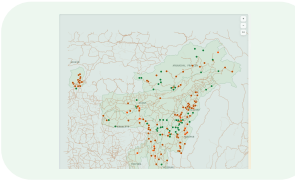
PROTOTYPE / PRODUCT

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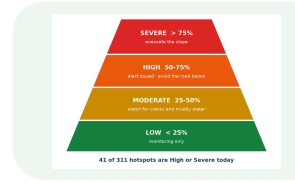
Screenshots of the live system — not mockups.











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KEY FEATURES / INNOVATION

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What makes your solution unique?

- Trained on 318 real, documented NER landslides (2007–2025) — not a generic global model.
- We caught a reporting bias: the model had learned “near roads” meant risk, not “steep and wet” — which under-warns remote villages.
- Terrain is read across a 300 m neighbourhood, not one point — coordinates usually mark where debris stopped, not the failure site.
- USGS slope-stability physics is combined with machine learning, not either alone.
- Works offline, and alerts render in 9 regional languages automatically.



What sets SPIREXA apart: region-specific training data, physics-informed features, and a caught-and-corrected reporting bias.

07

APPLICATIONS / IMPACT

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Where can it be used? What impact will it create?

- District administrations get a daily ranked watch-list instead of a phone call after the slope has already gone.
- Remote villages receive warnings in their own language, on basic phones, with no internet required.
- Road and power authorities see exactly which assets sit below an at-risk slope.
- Field officers get an offline reporting tool that syncs automatically when signal returns.



From district control rooms to individual field officers — one system, ranked by urgency, in a language people actually read.

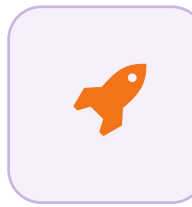
08

CURRENT STATUS & FUTURE PLAN

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Where are we now? What is next?

- Working today: satellite pipeline, ML model, GIS dashboard, prediction API, 9-language alerts, offline field app.
- In progress: an Arduino ground-sensor node feeding live rainfall and soil readings into the same dashboard.
- Planned: SMS gateway integration — message logic is ready, only a paid account is missing.
- Planned: a validation partnership with GSI or a state disaster management authority.



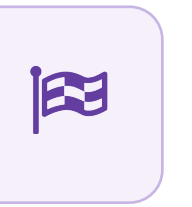
Software working

→



Sensor network

→



Official adoption

09

DEMO / SUPPORTING MATERIAL

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Demo video link:

Pitch deck link:

Product brochure:

Patent / paper:

Media coverage:

Demo requirement:

github.com/UtkarshCodecom/spirexa-ner — full source, live dashboard demo, and the Arduino sensor code

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KEY METRICS / ACHIEVEMENTS

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Users / customers:

Pilot / deployment:

Revenue / grant:

TRL / stage:

Awards:

Validation data:

0.933 ROC-AUC (5-fold cross-validated) · 87% of 318 real historical landslides flagged in advance · 311 hotspots monitored daily · TRL-4 (validated in a simulated / lab environment, not yet field-deployed)

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VISITOR NOTES / FOLLOW-UP

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Visitor query:

Interested person:

Organization:

Contact:

Follow-up action:

Remarks:

Use this space during exhibition for interaction notes, leads, and follow-up points

CONTACT & CONNECT

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GitHub:

github.com/UtkarshCodecom/spirexa-ner

LinkedIn:

Email:

yushkarsh@gmail.com

Phone:

Address:



LOOKING FOR

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Industry Collaboration

Investment / Funding

Mentorship

Technology Partnership

Pilot / Deployment Opportunities

Talent / Internship

Innovate. Ignite. Impact.

Building Ideas. Creating Impact. Empowering Tomorrow.

#SPIREXA